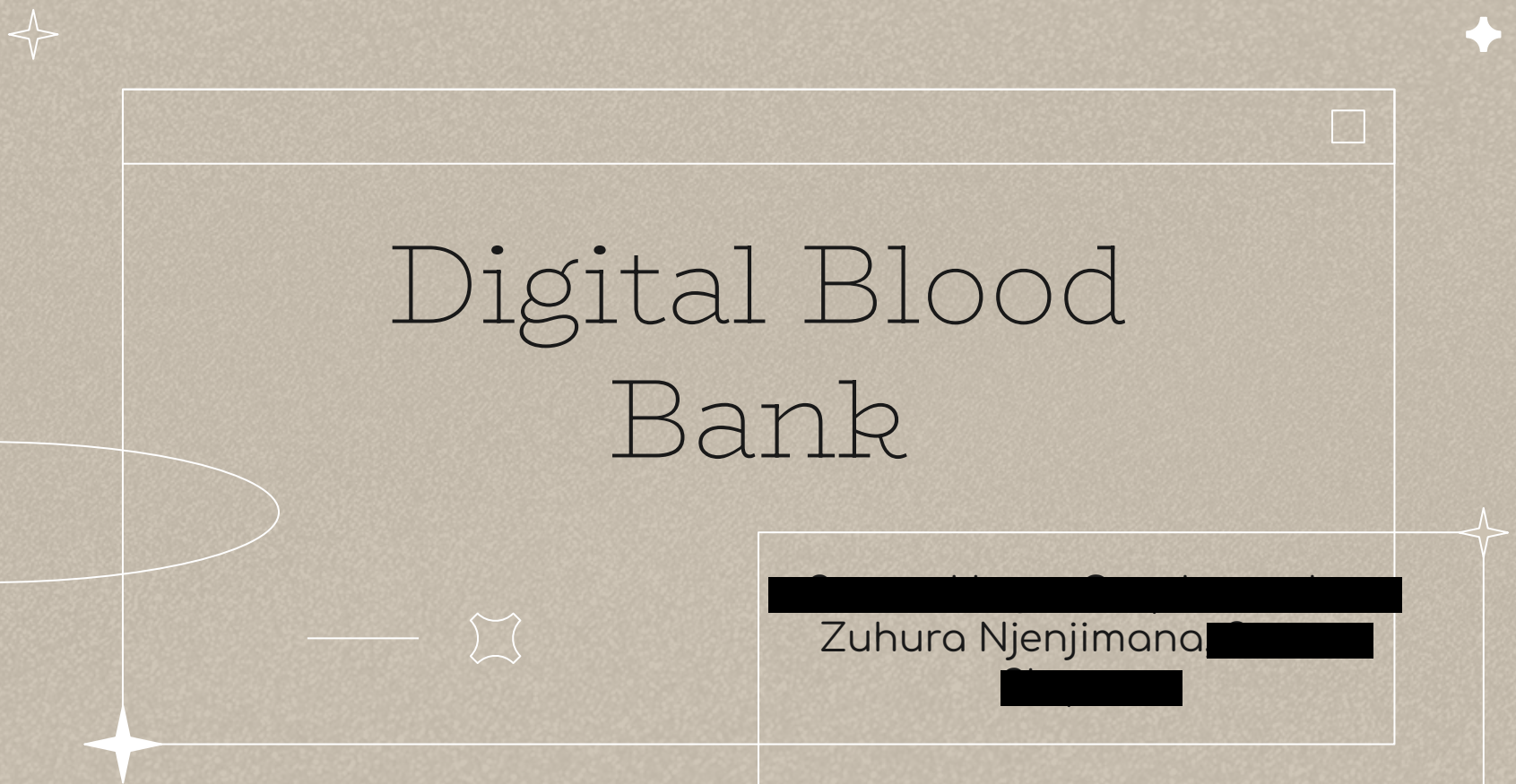




Digital Blood Bank

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Digital Blood Bank & Transfusion Matching System

This project demonstrates how MATLAB can be used to simulate a hospital blood bank system that selects the safest and most practical blood unit for a patient. Blood transfusions are critical in healthcare because they save lives during surgery, trauma, cancer treatment, anemia, and severe blood loss. However, incompatible blood can cause serious transfusion reactions, so accurate matching is essential.

The purpose of our project was to design a digital blood bank system that identifies the best available blood unit by using compatibility rules, patient history, urgency level, and inventory management. This creates a more realistic and useful healthcare model.

Biomedical Engineering Connection

This project strongly relates to biomedical engineering because it combines medicine, biology, and computer programming. Biomedical engineers develop systems and technologies that improve patient care, safety, and hospital efficiency.

Our project uses biological knowledge of ABO and Rh blood typing together with an algorithm that helps make transfusion decisions. It demonstrates how engineering can solve real clinical problems through automation and smart decision-making.



✧ Key Variables and Code Factors — ✧

1. Patient.bloodType :Stores the patient's blood type and determines which donor units are compatible.

2. Patient.urgency:A scale from 1 to 10. Higher urgency increases priority for faster blood access.

3.bloodDB.Quantity:Tracks how many blood units are available in inventory. This helps protect rare blood types and manage stock levels.

Additional Variables

bloodDB.DaysOld → Prefers fresher blood

bloodDB.Available → Checks if unit is in stock

Patient history → Includes transfusions, pregnancy, reactions, chronic illness

How the MATLAB Code Works

```
>> %% =====  
% BLOOD TRANSFUSION MATCHING SYSTEM (REWRITTEN + FIXED)  
% Compatibility verified and ranking corrected  
%% =====  
clc;  
clear;  
%%  
% BLOOD TYPE DATA  
%%  
Input = ["A+", "A-", "B+", "B-", "O+", "O-", "AB+", "AB-"];  
Receive = {  
    ["A+", "A-", "O+", "O-"]      % A+  
    ["A-", "O-"]                 % A-  
    ["B+", "B-", "O+", "O-"]      % B+  
    ["B-", "O-"]                 % B-  
    ["O+", "O-"]                 % O+  
    ["O-"]                       % O-  
    ["A+", "A-", "B+", "B-", "O+", "O-", "AB+", "AB-"] % AB+  
    ["AB-", "A-", "B-", "O-"]     % AB-  
};
```

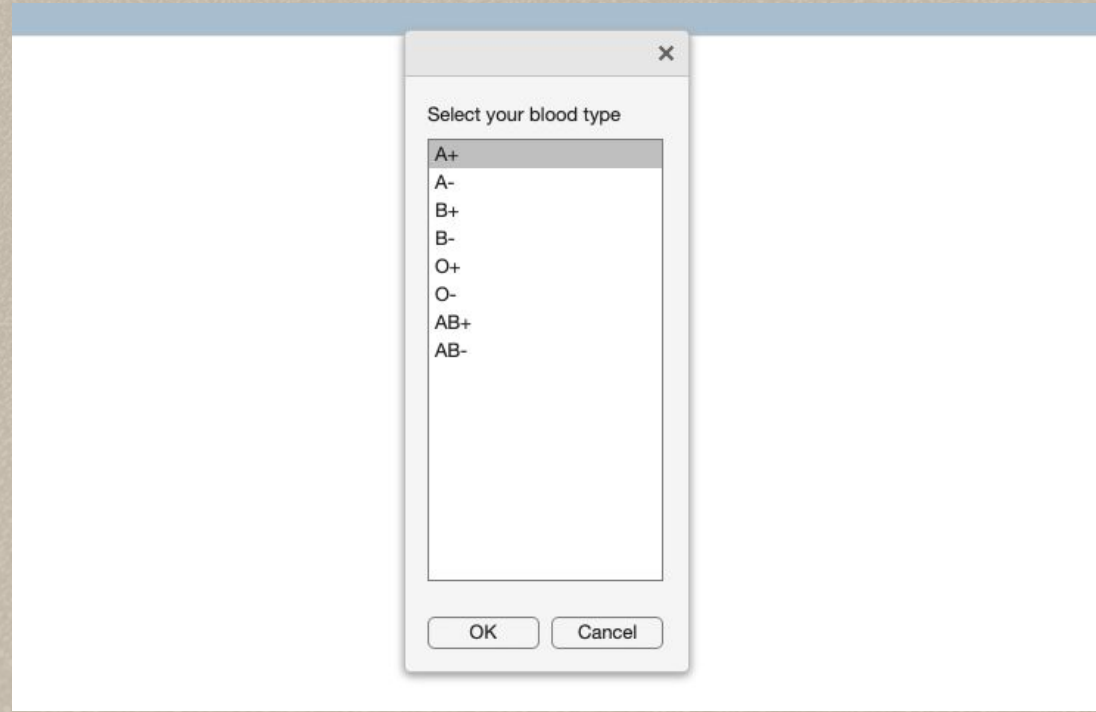
Green text: Organizational notes

Input: Lists all blood types used in application

Receive: Cell array that matches the person's blood type with options of other compatible blood types.

Demonstration / Output

In this example A+
was chosen as the
patient's blood type.



How the MATLAB Code Works

```
%% -----  
% PATIENT INPUT  
%% -----  
[indx,tf] = listdlg( ...  
    'PromptString','Select your blood type', ...  
    'SelectionMode','single', ...  
    'ListString',Input);  
if tf == 0  
    return;  
end  
conditionList = ["Normal","Sickle Cell","Thalassemia", ...  
    "Anemia","Cancer","Autoimmune Disorder"];  
[condIdx,tf2] = listdlg( ...  
    'PromptString','Select condition', ...  
    'SelectionMode','single', ...  
    'ListString',conditionList);  
if tf2 == 0  
    return;  
end  
urgencyInput = inputdlg('Enter urgency level (1-10):');  
historyInput = inputdlg('Number of past transfusions:');  
pregnancyQ = questdlg('History of pregnancy?', ...  
    'Patient History','Yes','No','No');  
reactionQ = questdlg('Previous transfusion reaction?', ...  
    'Patient History','Yes','No','No');  
chronicQ = questdlg('Any chronic illness?', ...  
    'Patient History','Yes','No','No');  
conditionsInput = inputdlg( ...  
    'List pre-existing conditions (comma separated):');
```

[indx, tf]=listdlg: Displays the blood type Input list as a clickable menu.

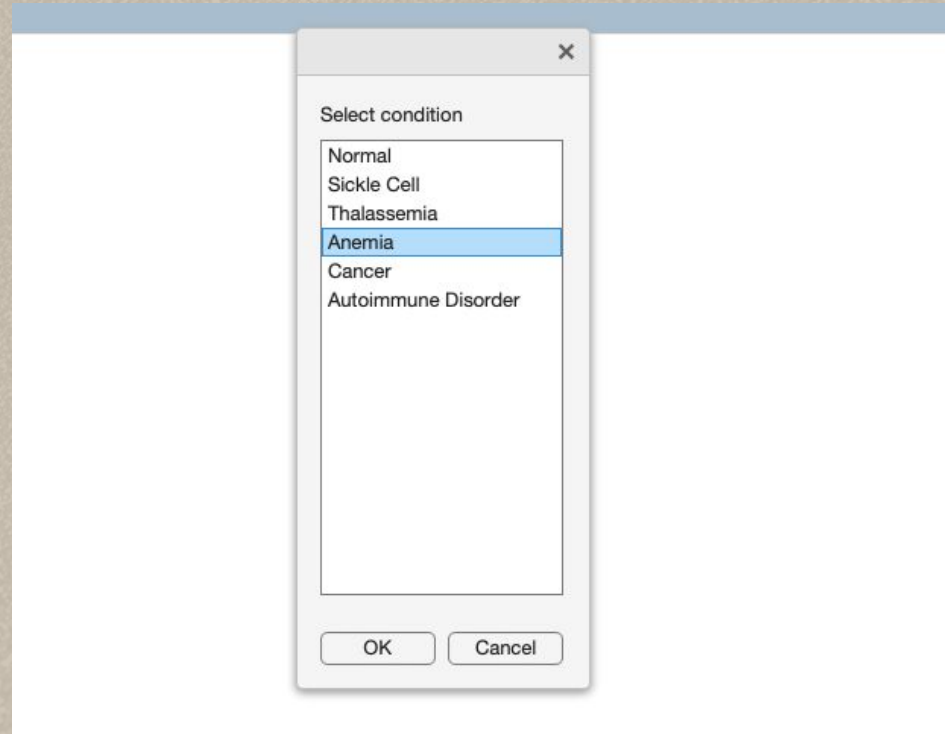
conditionList: Shows second menu with possible medical conditions.

inputdlg: Allows the user to type in information relevant to their care. Purple text in parenthesis is the question asked.

questdlg: Allows the user to answer yes/no questions by clicking a 'Yes' button or a 'No' button.

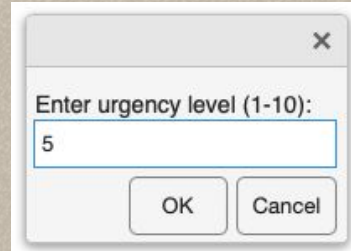
Demonstration / Output

Then the condition chosen
was Anemia



Demonstration / Output

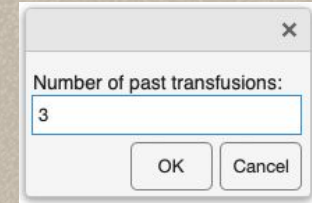
The number chosen was 5. The number of past transfusions chosen is 3. "No" for Pregnancy history and "No" for previous transfusion reaction.



Enter urgency level (1-10):

5

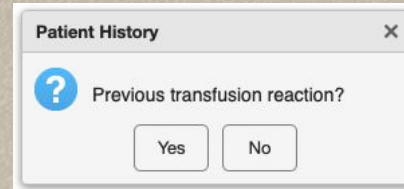
OK Cancel



Number of past transfusions:

3

OK Cancel



Patient History

? Previous transfusion reaction?

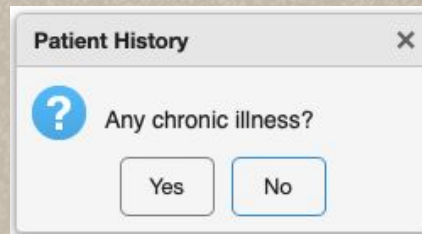
Yes No



Patient History

? History of pregnancy?

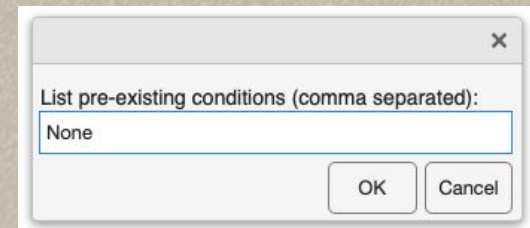
Yes No



Patient History

? Any chronic illness?

Yes No



List pre-existing conditions (comma separated):

None

OK Cancel

How the MATLAB Code Works

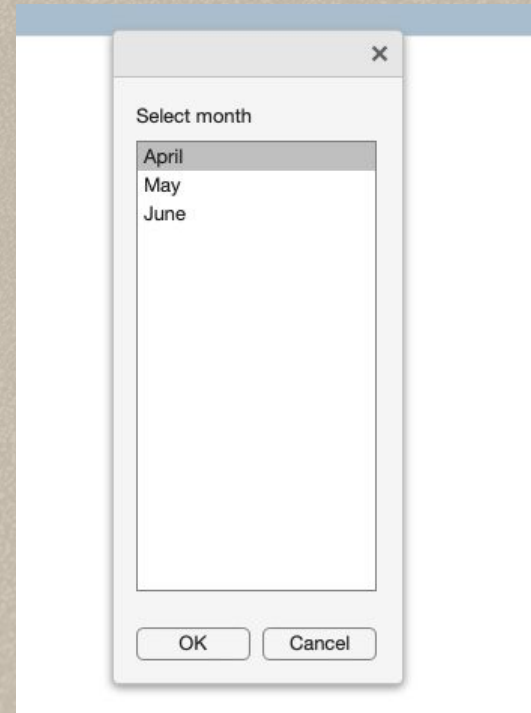
```
>> %% =====  
% APPOINTMENT BOOKING  
%% =====  
months = ["April","May","June"];  
[monthIdx,tfm] = listdlg( ...  
    'PromptString','Select month', ...  
    'SelectionMode','single', ...  
    'ListString',months);  
if tfm == 0  
    return;  
end  
days = string(1:31);  
[dayIdx,tfd] = listdlg( ...  
    'PromptString','Select day', ...  
    'SelectionMode','single', ...  
    'ListString',days);  
if tfd == 0  
    return;  
end
```

Months: creates a list for the months appointments are available

if tfm == 0: if a user x's out of the menu, the code will stop everything.

Demonstration / Output

The month chosen
for appointment
booking was April.



How the MATLAB Code Works

```
>> days = string(1:31);  
[dayIdx,tfd] = listdlg( ...  
    'PromptString','Select day', ...  
    'SelectionMode','single', ...  
    'ListString',days);  
if tfd == 0  
    return;  
end  
unavailableDays = ["15","16"];  
if any(strcmp(days(dayIdx), unavailableDays))  
    disp("Selected date unavailable. Restart booking.")  
    return;  
end
```

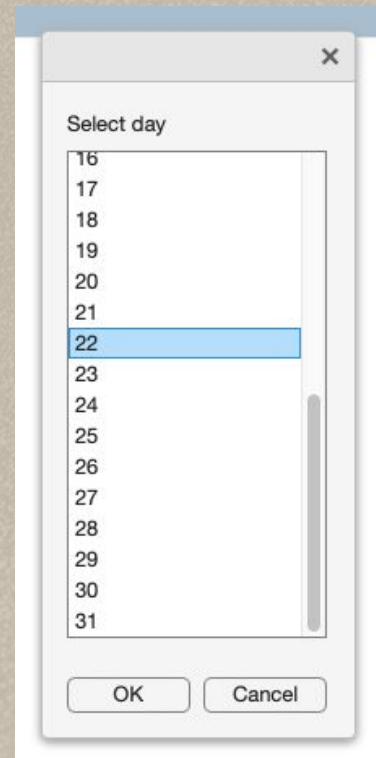
unavailableDays: Tells the user which days are unavailable to book an appointment

strcmp: Compares the user's chosen appointment day to the day's availability

return: Prevents the code from moving forward with an invalid date, forcing the user to start over.

Demonstration / Output

The day 22nd of
April.



How the MATLAB Code Works

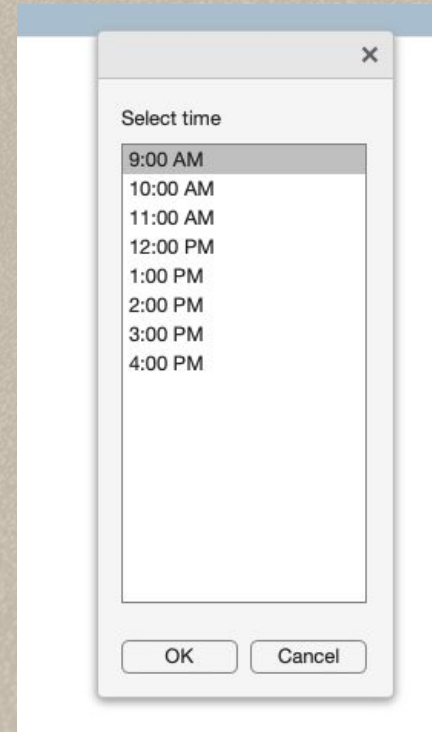
```
>> times = ["9:00 AM","10:00 AM","11:00 AM","12:00 PM", ...  
            "1:00 PM","2:00 PM","3:00 PM","4:00 PM"];  
[timeIdx,tft] = listdlg( ...  
    'PromptString','Select time', ...  
    'SelectionMode','single', ...  
    'ListString',times);  
if tft == 0  
    return;  
end  
slotsAvailable = randi([0,3]);  
if slotsAvailable == 0  
    disp("No slots available. Restart booking.")  
    return;  
end
```

times: Creates a list of strings representing the available appointment times from 9:00 AM to 4:00 PM

randi: Generates a random integer between 0 and 3. If the random number generator picks 0, the code assumes that specific time slot is full.

Demonstration / Output

The time chosen
was 9:00 am sharp.



How the MATLAB Code Works

```
>> %% -----  
%% CONFIRMATION  
%% -----  
disp(" ")  
disp("----- APPOINTMENT CONFIRMED -----")  
disp("Date: " + months(monthIdx) + " " + days(dayIdx))  
disp("Time: " + times(timeIdx))  
disp("Blood Reserved: " + bestMatch)
```

disp: Displays all the text in purple to the user. Gives them all of the information they would need to know for their appointment.

Demonstration / Final Output

This displays detailed and summarized info of patient's history and appointment for Blood bank to keep note of until appointment date.

```
---- PATIENT INFO ----  
Blood type: A+  
Condition: Anemia  
Urgency: 5  
Compatible donor types: A+, A-, O+, O-  
  
---- MATCH RESULT ----  
Best Blood Unit: A+  
Unit ID: 101  
Storage Location: Fridge A  
Remaining Quantity: 2  
Match Score: 180  
  
---- MATCH RESULT ----  
Best available blood unit: A+  
Match score: 180  
Compatibility Rating: Excellent Match  
Warning: Higher antibody risk  
  
---- APPOINTMENT CONFIRMED ----  
Date: April 22  
Time: 9:00 AM  
Blood Reserved: A+
```

Importance & Real-World Application



Transfusion safety depends on accurate compatibility testing and effective blood inventory management. Errors in blood matching can lead to serious adverse reactions, which is why digital decision-support systems can help reduce human error, improve efficiency, and support faster responses during emergency situations.

A system like this could be applied in hospitals, blood banks, trauma centers, surgical departments, and outpatient clinics. It would be especially useful for patients who require repeated transfusions, such as individuals with thalassemia, anemia, or sickle cell disease.





Future Improvements & Conclusion

Real hospital blood bank data to test the program with realistic patient and inventory information

Interactive graphs and dashboards for clearer visualization of donor rankings, stock levels, and appointments

Improved graphical user interface (GUI) for a more professional and user-friendly experience

Advanced antigen matching beyond ABO and Rh factors for higher-risk patients

Machine learning models to predict blood shortages and demand trends

Integration with hospital databases or electronic health records for real-time clinical use

Conclusion



This MATLAB project demonstrates how biomedical engineering can improve healthcare through the integration of programming, biological principles, and decision-making systems. It provides a structured model for safer transfusions, improved inventory control, and faster clinical decision-making.

By combining compatibility logic with patient-specific and inventory-based factors, the system reflects a practical approach to real-world hospital workflows and highlights the role of engineering in modern healthcare.



References

Dunbar, N. M. (2020). Does ABO and RhD matching matter for platelet transfusion? *Hematology*. <https://doi.org/10.1182/hematology.2020000135>

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